

# MAUVE-MUSE: When Metallicity Follows or Fights Star Formation

## A Mass Dependent Inversion in Virgo Spirals

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### INTRODUCTION

Resolved Mass-Metallicity Relation (rMZR) probes the balance of **metal enrichment vs dilution** in discs.

**Key question:** Does gas-phase metallicity ( $Z_{gas}$ ) **correlate or anti-correlate** with Star formation rate (SFR) surface density ( $\Sigma_{SFR}$ ) at fixed stellar mass surface density ( $\Sigma_*$ ), and does the sign changes across discs?

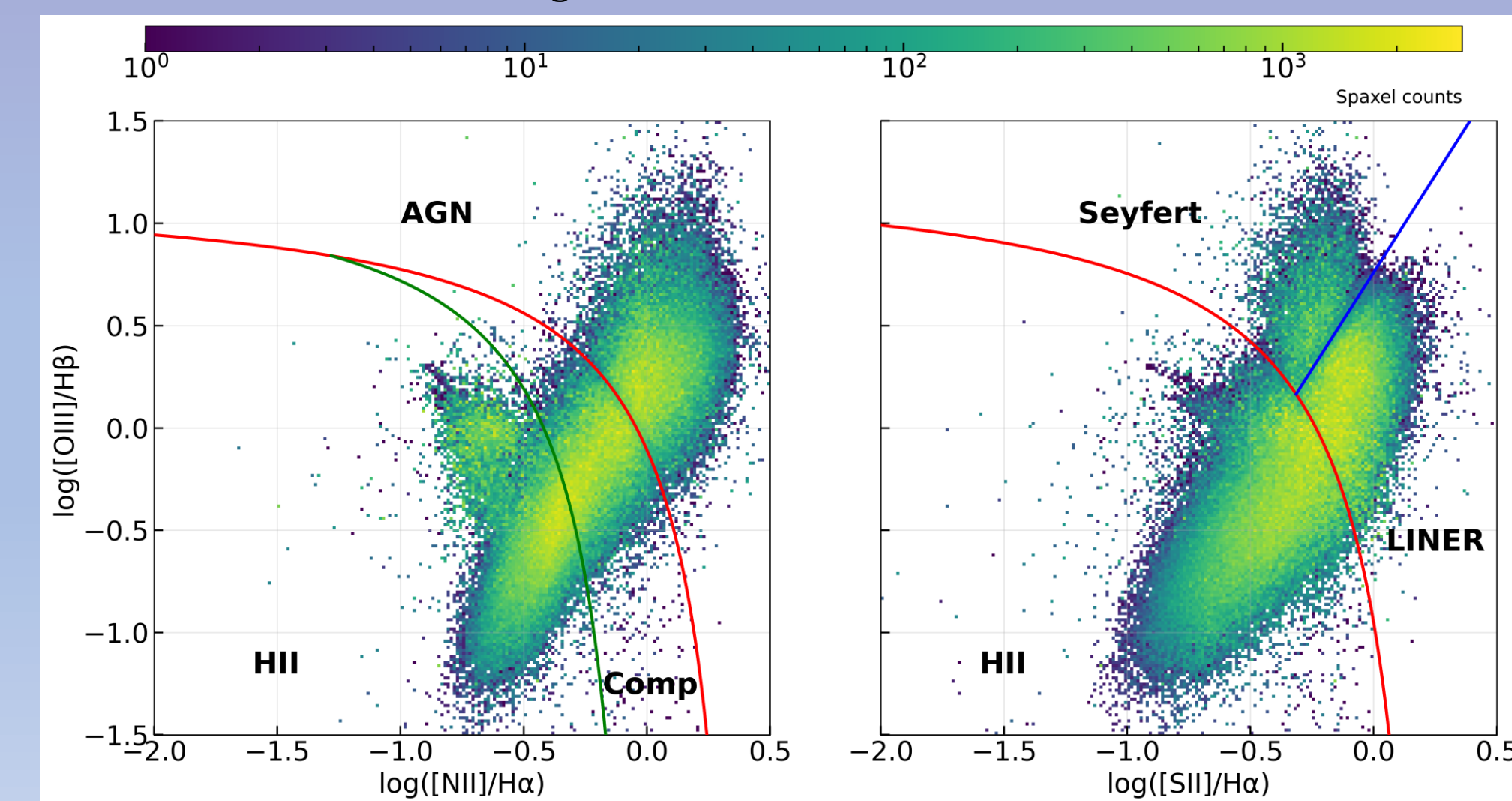
### DATA & METHODS

Data: MAUVE-MUSE (Virgo, 100 pc)

- Instrument:** VLT/MUSE wide-field IFU (optical IFS), delivering stellar mass and emission-line maps on  $\sim 100$  pc scales in Virgo spirals.
- Sample:** **14 Virgo spirals** (MAUVE-MUSE v2 internal release); survey observations are **ongoing**.
- Working set:** fiducial ensemble uses **13 galaxies** (excluding the strong outlier **NGC 4383**).
- Resolved Maps used (co-spatial spaxels):**  $\Sigma_*$ ,  $\Sigma_{SFR}$  and  $Z_{gas}$ .

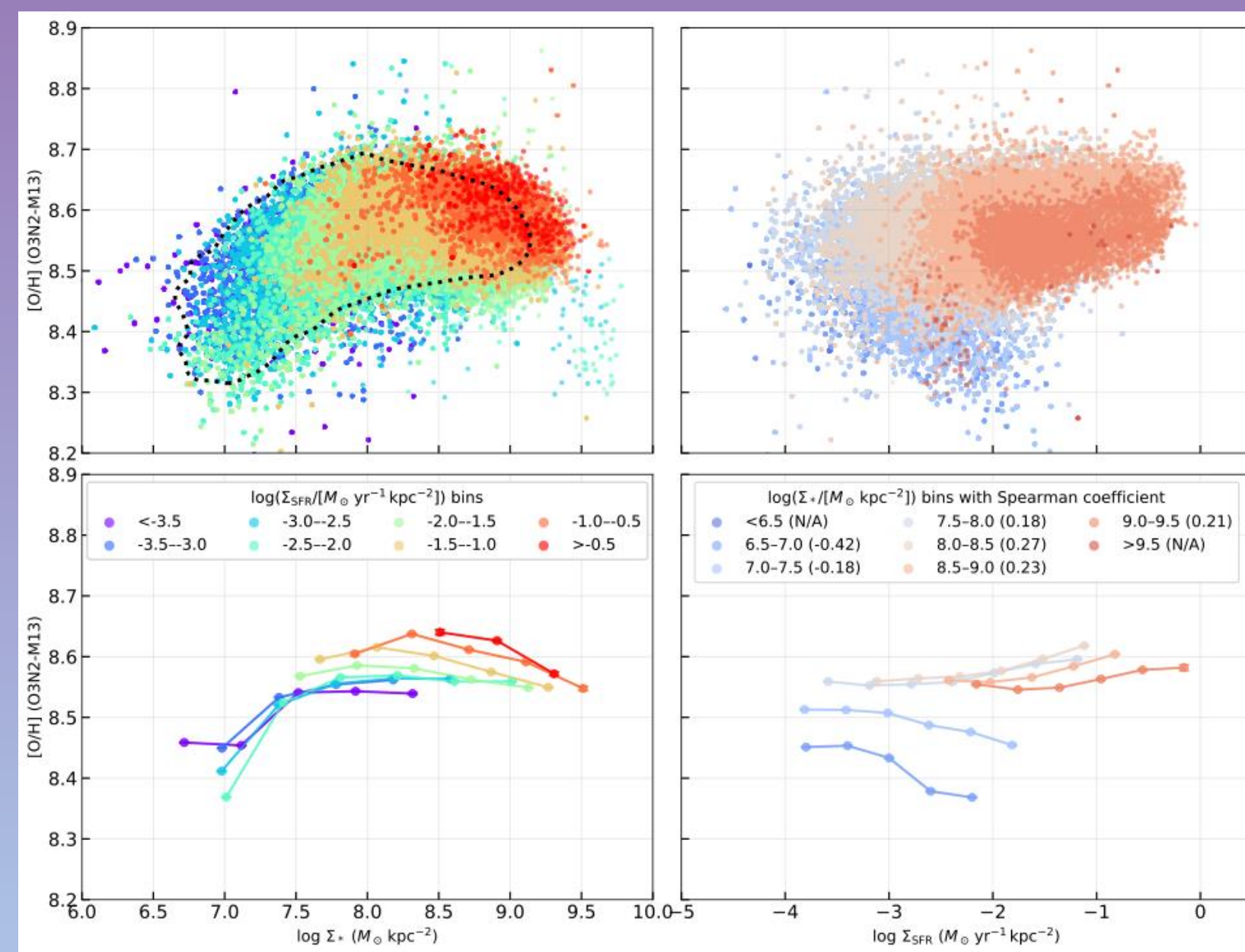
Method: isolate star-forming spaxel and measure  $Z_{gas}$

- Star-Forming selection:** keep H II spaxels using [N II] and [S II] BPT selection diagnostic cuts
- Calibration:** fiducial  $Z_{gas}$  from **O3N2-M13** (Marino+2013) using strong lines (dust-corrected H $\alpha$  H $\beta$ , [O III], [N II])
- $\Delta Z_{gas} - \Delta \Sigma_{SFR}$  coupling measurement: remove primary mass-dependent trends, then measure residual coupling  $\rho(\Sigma_*) = \text{corr}(\Delta Z_{gas}, \Delta \log \Sigma_{SFR})|_{\Sigma_*}$

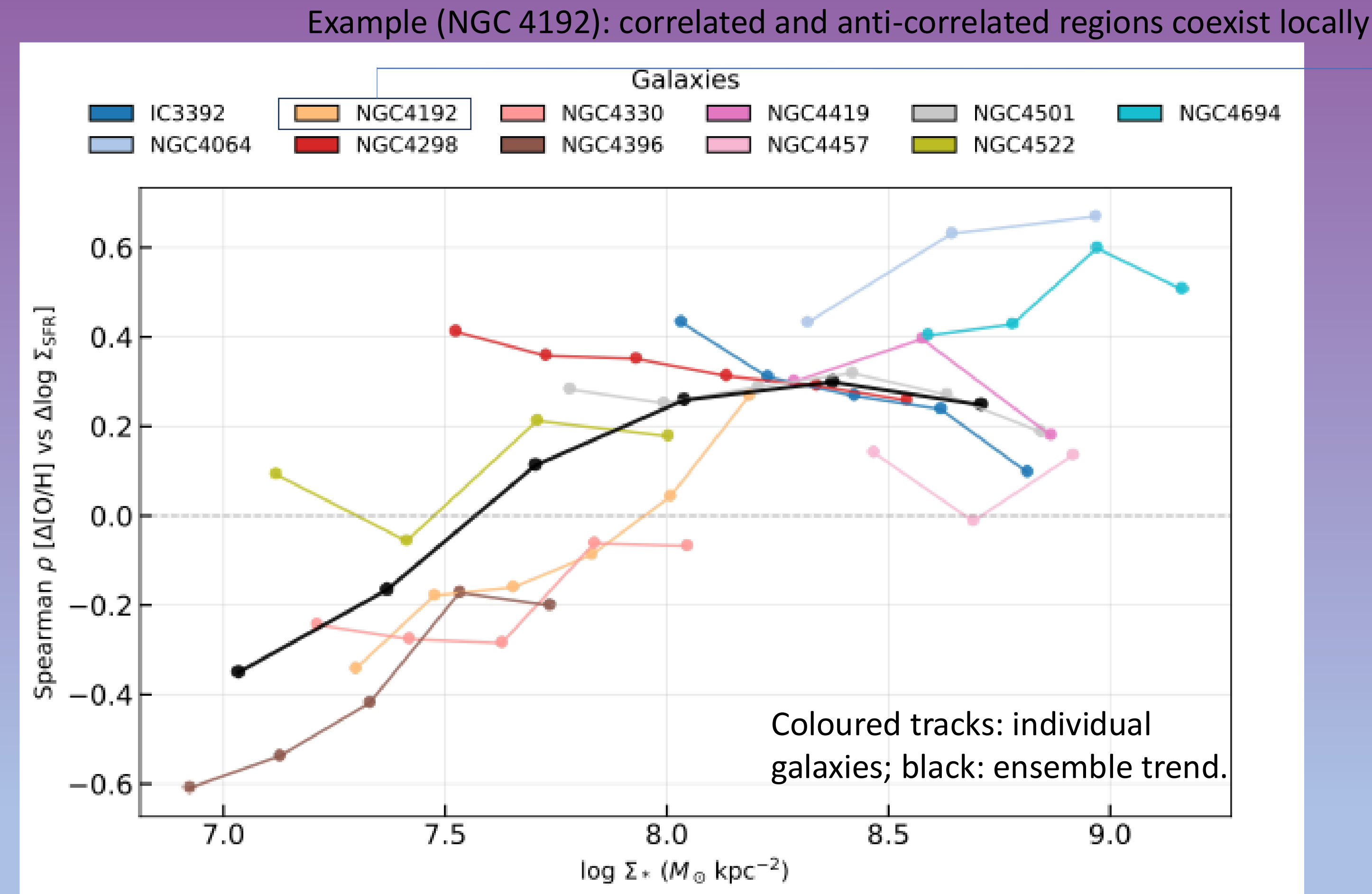


### RESULTS

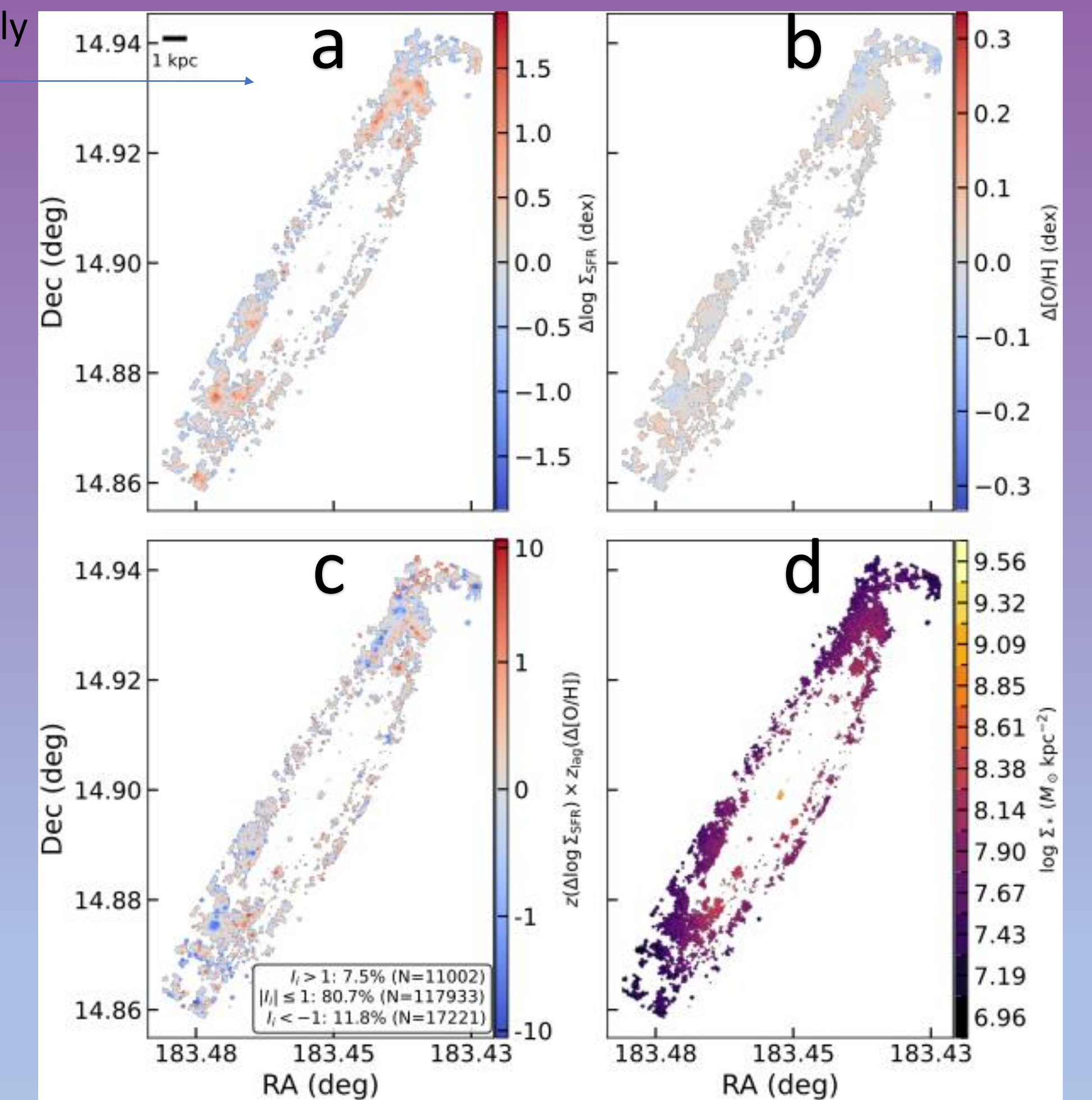
rMZR is well-defined: the *secondary*  $\Delta Z_{gas} - \Delta \Sigma_{SFR}$  relation depends on  $\Sigma_*$



**Left:** rMZR:  $Z_{gas}$  rises with  $\Sigma_*$ ; colour shows  $\Sigma_{SFR}$   
**Right:** within narrow  $\Sigma_*$  bins, the  $Z_{gas} - \Sigma_{SFR}$  trend changes sign: **low  $\Sigma_*$  anti-correlation**  $\rightarrow$  **high  $\Sigma_*$  positive correlation**.



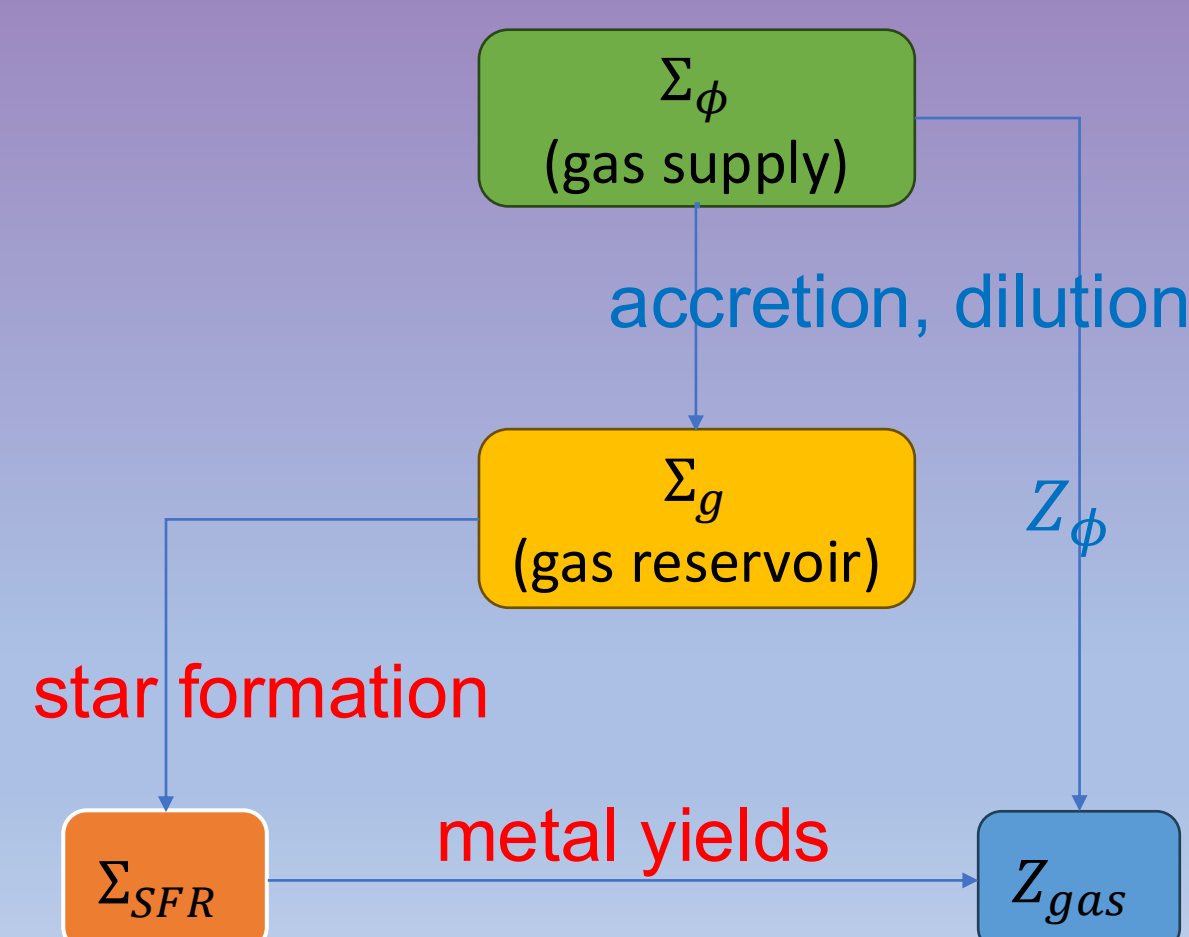
We compute a Spearman rank correlation between residual  $\Delta Z_{gas} - \Delta \Sigma_{SFR}$ . Across galaxies,  $\Sigma_*$  transitions smoothly from **negative** (gas-supply-dominated) to **positive** (star-formation-dominated), with an inversion point at  $\log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \approx 7.5 - 8.0$



a,b,d panels:  $\Sigma_{SFR}$ ,  $Z_{gas}$  and  $\Sigma_*$  maps. Panel c: A local Moran-like index quantifies neighbourhood co-variation between  $\Delta Z_{gas}$  and  $\Delta \Sigma_{SFR}$ . **red = concordant patches, blue = discordant patches**.

### INTERPRETATION: LOCAL GAS-REGULATOR

$$\frac{d}{dt} \Sigma_g = \Sigma_\phi - (1 - R) \Sigma_{SFR} - \Sigma_{out}; \quad \frac{d}{dt} Z_{gas} = \frac{1}{\Sigma_g} [y(1 - R) \Sigma_{SFR} + (Z_\phi - Z_{gas}) \Sigma_\phi]$$



A spatially resolved *bathtub* model to describe the local gas regulation.

At fixed  $\Sigma_*$ :

$\Delta Z_{gas} \propto \Delta \log \Sigma_{SFR} - \Delta \log \Sigma_\phi$   
**Star-formation-driven variability** dominates  $\Rightarrow \uparrow \Sigma_{SFR}$  with  $Z_\phi$  dilution  $\Rightarrow \downarrow Z_{gas}$ .  
**Gas-supply-driven variability** dominates  $\Rightarrow \downarrow \Sigma_{SFR}$  with metal yields  $\Rightarrow \uparrow Z_{gas}$ .

### CONCLUSIONS

- Detect** a mass-dependent inversion in residual  $\Delta Z_{gas} - \Delta \Sigma_{SFR}$  coupling
- Show** that this flip-sign is spatially varying in the local Moran-like map
- Interpret** as a transition from **gas supply/dilution-dominated** to **star formation/enrichment-dominated** regulation
- Implication:** this resolved trends unify with integrated metallicity-SFR behaviour.

### ROBUSTNESS

- The **sign flip persists** across O3N2-based calibrations;
- HII+Comp BPT selection **keeps the same  $\Sigma_*$  transition**;
- Metallicity outlier, NGC4383, points out a **potential breakdown** in outflows/extraplanar components and/or strong, recent gas-supply perturbations.

### CONTACT INFORMATION

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